

Master Thesis Presentation

Good afternoon. My name is Heykel Khadhraoui, and today I will present my master's thesis on real-time impedance control for a seven-degree-of-freedom collaborative robot.

The intended application is surface grinding with a rectangular tool. My work focuses on contact entry: how the tool can rotate towards alignment with a planar surface through compliant contact.

I will begin with the problem that motivates this behaviour.

Motivation

The illustration shows the basic problem. When a rectangular tool approaches a surface with a small tilt, it touches along an edge or at a corner before the whole face makes contact. Small differences in surface placement, tool mounting, and robot pose can produce this situation.

A rigid pose controller tries to maintain the commanded orientation. It therefore resists the corrective rotation that contact could generate, and the contact loads can increase.

The idea in this thesis is to maintain the normal press while allowing the tool to rotate through its interaction with the surface. Here, passive alignment means that contact produces the rotation, rather than an additional prescribed alignment trajectory.

The main question is: how do stiffness and the centre of compliance affect this response? Stiffness determines resistance to motion. The centre of compliance is a virtual reference point that can change the rotational effect of the press.

The presentation follows these two ideas, from the controller to the experiments.

Overview

First, I will explain Cartesian impedance and the centre of compliance. Then I will present the contact experiments and their results. A separate study examines null-space control. I will finish with the conclusion and future work.

I will start with the tool pose and the errors used by the controller.

Cartesian impedance wrench

Cartesian wrench means force and moment together. The controller generates these commands by comparing the desired and measured tool pose. The tool centre point, or TCP, is the controlled reference, and its position and orientation are expressed relative to the robot base frame.

The left column concerns position and force. Here, p_d is the desired position and p_{EE} is the measured position. Their difference is the position error.

$$e_p = p_d - p_{EE}.$$

The commanded force combines a spring contribution from this error and a damping contribution from the difference between desired and measured linear velocity.

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE}).$$

The right column applies the same idea to orientation and moment. R_d is the desired orientation, and R_{EE} is the measured orientation. The error e_R describes the rotational correction from measured to desired orientation.

$$m = K_R e_R + D_R (\omega_d - \omega_{EE}).$$

The symbol key collects the common terms. K_p and K_R are translational and rotational stiffness; D_p and D_R are the corresponding damping matrices. A dot over position denotes linear velocity, and omega denotes angular velocity. The indices d and EE mean desired and measured end effector.

Stiffness resists pose errors, while damping reduces oscillation. The next slide shows how the force and moment reach the robot joints.

Real-time control

The loop shows how the controller closes the feedback path. It calculates the pose error from the desired and measured tool pose, generates the impedance wrench, and maps that wrench to joint torques. The robot moves, and its measured pose returns to the next update.

The equation gives the torque mapping. F collects the commanded force and moment. The Jacobian transpose maps this Cartesian wrench to the Cartesian task's joint-torque contribution.

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^T(q)F.$$

Here, q denotes the measured joint positions. The geometric Jacobian, J , depends on this configuration and relates joint motion to Cartesian motion. Its transpose maps the six wrench components to seven joint-torque components.

I implemented this feedback controller in C++ using libfranka. Its nominal torque-control frequency is one kilohertz, or one update every millisecond.

The next slide explains how the compliant behaviour is chosen relative to the surface.

Surface-relative compliance

These directions are defined relative to the configured surface. The arrow n_s is the surface normal: it is perpendicular to the plane and points outward. Pressing towards the surface therefore follows the opposite direction.

The other two arrows, t_1 and t_2 , are perpendicular tangent directions lying in the plane. Together, these three directions form the surface frame.

For translation, I use softer behaviour along the normal to allow compliant pressing. Along the two tangents, higher stiffness helps retain the tool's position on the surface.

For rotation, low stiffness about the two tangents allows rotations that bring the tool face towards alignment with the plane. Higher stiffness about the normal resists rotation in the plane and maintains the tool's in-plane orientation.

This distinction is important: translating along a tangent and rotating about that tangent are different motions. The controller can assign a different stiffness to each.

Besides choosing these directions, I can also shift the point about which the impedance is defined.

Centre of compliance (CoC)

The tool centre point, or TCP, is the controlled reference attached to the end effector. The centre of compliance is the virtual point about which the spring–damper behaviour is defined. It is a controller setting, not a physical hinge.

Its position is written as p_c . The vector r_c points from the TCP to this virtual centre. Moving the centre away from the TCP couples translation and rotation: a translational error can then also contribute to a commanded moment.

The equation summarises this effect. The complete commanded moment m contains the ordinary rotational spring–damper moment m_R , plus the additional contribution r_c cross f . Here, f is the commanded force introduced earlier.

The diagrams isolate the normal part of that force, written as f_n . The tool tilt and normal press are the same in both panels. Only the side of the tangential displacement changes. This reverses the additional moment, as indicated by the curved arrows.

For a given tilt, one side can support the aligning rotation, while the opposite side can oppose it. If the centre is at the TCP, r_c is zero and this additional contribution vanishes. The ordinary rotational stiffness and damping still remain active.

With the stiffness directions and compliance centre defined, I will now describe the contact experiments.

$$m = m_R + r_c \times f.$$

Experimental procedure

The contact study used a rectangular tool measuring forty by one hundred and twenty millimetres. Each experiment followed the same sequence.

First, the robot oriented the tool and approached the surface. During the five-second Contact Establishment phase, the selected tool-point reference advanced towards the surface. Contact restricted translation, so the resulting position error generated the pressing force through the impedance controller. The robot then entered the pre-grinding hold.

The desired orientation remained fixed at the measured contact-entry orientation. The controller prescribed no alignment rotation; the observed rotation developed through compliant contact.

Case A established the baseline with the centre of compliance at the TCP. Cases B and C varied rotational stiffness and tangential translational stiffness. Case D varied the tangential centre-of-compliance position. Within each comparison, the other settings were kept unchanged.

The contact study contained nineteen parameter settings, each repeated three times, giving fifty-seven trials. The quantitative comparisons concern contact entry under fixed conditions.

Before showing the results, I will define the two angular quantities used to evaluate the contact-entry response.

Angular quantities

Two angular quantities describe the contact-entry response. Theta achieved about t one is the achieved entry tilt relative to the configured surface reference. Gamma about t one measures the contact response during Contact Establishment.

Gamma is defined by the rotation from the measured end orientation back to the held entry reference. Its sign is therefore opposite to the physical start-to-end rotation. Under this convention, an aligning response has the same sign as the entry tilt.

The response describes how much the tool rotated during contact. It does not directly measure the remaining alignment error, so a larger response does not by itself establish more accurate final alignment.

Both quantities are calculated from measured end-effector orientation. They do not provide an independent measurement of the physical tool–surface angle.

$$\theta_{\text{ach},t_1} : \text{achieved entry tilt}, \quad \gamma_{t_1} : \text{contact response}.$$

With these definitions, we can read the baseline results.

Baseline contact response

Here, the centre of compliance is at the TCP. The horizontal axis lists three entry conditions: nominal zero offset, positive tilt, and negative tilt. The positive and negative achieved entry angles are approximately plus nine point three and minus nine point four degrees. The nominal-zero label identifies a test condition; it does not assert a measured angle of exactly zero.

The vertical axis shows the contact response. Each value is the average of three trials. The positive-tilt condition produced approximately plus seven point six degrees of response, and the negative-tilt condition approximately minus ten point eight degrees. Both signs indicate rotation in the aligning direction, although their magnitudes differ. The nominal-zero condition produced a small negative response of approximately one degree.

These results establish that contact can generate aligning rotation for either entry-tilt sign with the compliance centre at the TCP. They describe the angular response and do not independently establish the final physical tool–surface angle.

I will now show how stiffness changes this response.

Effect of stiffness

The left plot shows the effect of rotational stiffness. K_{R,t_1} means the stiffness resisting rotation about the first surface tangent. These trials used the positive-tilt condition, with the other settings held constant.

Increasing this stiffness from five to fifty newton metres per radian reduced the response from approximately seven point six to two point seven degrees: a reduction of about sixty-four percent.

This agrees with the spring interpretation: greater rotational stiffness opposes more of the rotation produced by contact.

The right plot changes a different quantity, K_{p,t_2} . This is translational stiffness along the second tangent, so it resists sliding along the surface rather than rotation.

Changing it from three hundred to two thousand newtons per metre produced a response span of only three hundredths of a degree, or about zero point four percent of the baseline response.

Within the tested range, the alignment response was therefore much more sensitive to rotational stiffness than to tangential translational stiffness.

The next comparison changes the compliance-centre position while retaining the stiffness settings.

Effect of CoC position

The horizontal axis shows r_{c,t_2} , the centre-of-compliance displacement along the second surface tangent. The vertical axis remains the contact response about the first tangent. The curves are labelled by positive and negative achieved entry tilt. The error bars show one sample standard deviation across three trials.

For the positive-tilt condition, a displacement of plus forty millimetres produced approximately seven point nine degrees of response, compared with seven point six degrees at the TCP. Moving the centre to minus forty millimetres reduced the response to approximately zero point two degrees.

For the negative-tilt condition, the supporting side reverses. A negative displacement supports the negative response, while a positive displacement reduces its magnitude towards zero. Moving towards zero here means less rotation, not an improvement in alignment.

Relative to the TCP-centred case, the supporting side increased the response magnitude by about four percent. The opposing side reduced it by approximately ninety-eight to ninety-nine percent. The main effect was therefore the selection of a preferred rotational direction.

The time histories connect this directional effect to the interaction moment.

Contact response and interaction moment

These plots compare three centre positions for the positive-entry-tilt condition. Black is minus forty millimetres, red is the TCP-centred case, and blue is plus forty millimetres. The horizontal axis is time during Contact Establishment.

The left plot shows the contact response. In these representative traces, the supporting displacement reaches its approximately steady response at about two point three seconds, compared with about three point six seconds at the TCP. This is about one point three seconds earlier. This comparison describes the displayed traces, rather than a general timing result.

The right plot shows the robot's estimated interaction moment about the first surface tangent, referenced to the TCP. This is an estimate of the physical interaction, whereas the earlier compliance-centre equation described the moment commanded by the controller. The estimate comes from the robot's model-based external-wrench estimator, rather than an independent force–torque sensor.

Reversing the CoC displacement reverses the interaction moment. The final normal-force estimates are approximately minus eighty-three, minus seventy-nine, and minus seventy-eight newtons at minus forty millimetres, the TCP, and plus forty millimetres, respectively. These are means over the final second and show similar normal pressing levels. The complete three-panel comparison is available in the backup slides.

These observations connect the direction-dependent response to the interaction moment. The next video shows the contact sequence and adaptation during the pre-grinding hold.

Contact demonstration

This video shows the robot picking up the tool, approaching the surface, and establishing contact. It then enters the pre-grinding hold.

During the hold, I manually change the surface orientation. Normal pressing remains active, and the compliant tool adapts to the changed surface. This demonstrates adaptation during the hold, rather than the fixed-condition contact-entry comparisons in the result plots.

Here, the centre of compliance is at the TCP. The surface contact moment drives the rotation. The controller's rotational spring and damper remain active, but the additional moment from a displaced compliance centre is zero. The desired orientation remains fixed.

This completes the contact study. I will now turn to the separate study of redundant joint motion while the tool pose is held.

Secondary study: null-space motion

The contact study concerned motion of the tool. This secondary study concerns coordinated joint motion while the Cartesian controller holds the tool pose.

The robot has seven joint coordinates, while the tool pose has six degrees of freedom: three for position and three for orientation. At a nonsingular configuration, one independent direction of redundant motion remains. Several joints generally move together; the freedom is not assigned to one particular joint.

Projected damping opposes redundant joint velocity. Configuration conditioning instead steers the joints towards better motion capability and can initiate motion from rest. A separate pose-hold experiment evaluated these contributions individually under a repeatable, internally commanded disturbance.

The figure on the left shows cumulative projected null-space motion, E_N . Motion in both directions contributes to this accumulated quantity. The black curve has no null-space torque; the red curve uses projected damping. The curves are averages of three trials, and the shaded bands show one sample standard deviation on either side.

Damping reduced the accumulated motion from approximately seven point six to five point seven degrees, a reduction of about twenty-five percent.

The table shows a different measure: the mean net projected displacement along the reference disturbance-induced direction during the disturbance interval. It compares the inactive, damping, and two conditioning settings. The mean decreases from about seven point five degrees without null-space torque to five point six degrees with damping. For conditioning gains of one point five and two, the means are zero point zero one five degrees and minus zero point zero zero six degrees. Both are close to zero.

The graph shows total accumulated motion; the table shows the net displacement after motion in opposite directions cancels. A small net value does not mean that the joints did not move. The stronger conditioning setting produced more back-and-forth motion despite its small net displacement.

Across the study, the largest measured TCP position error was one point three zero four millimetres, below the two-millimetre criterion. The next video illustrates the three controller settings.

Disturbance demonstration

This video shows three responses to a disturbance while the Cartesian controller holds the tool pose.

First, no null-space torque is active, and the disturbance produces internal joint motion. Next, projected damping is active. Its torque opposes the null-space joint velocity and reduces the motion.

Finally, conditioning is active. This torque adjusts the joint configuration to improve its motion capability. It can generate internal motion while steering the configuration, so its role differs from damping.

Across the three settings, the focus is on how the arm configuration changes while the tool remains approximately in place. The preceding slide separates the total accumulated motion from the net projected displacement.

I will now finish with the main conclusions and proposed future work.

Conclusion

The implemented Cartesian impedance controller enabled passive contact-induced rotation of the rectangular tool. Higher rotational stiffness reduced the angular response. A tangential centre-of-compliance displacement selected a preferred rotational direction.

For future work, I propose selecting the CoC displacement direction from the measured entry tilt and returning the centre to the TCP after alignment. Direct tool-angle measurement and sustained grinding trials would extend the present contact-entry evaluation.

The main message is that the compliance-centre position should be chosen with respect to the required alignment direction. Thank you for your attention. I welcome your questions.

Commanded and estimated wrench

This supplementary comparison examines the commanded and model-estimated wrench during one contact trial. Wrench means the combined force and moment. Here, F_n is the normal-force component, and M_{t_1} is the moment about the first surface tangent.

Both quantities are compared using the same surface axes and TCP reference point. The model-estimated signal is the interaction estimate available from the robot.

The shaded final interval is stationary. Over that interval, the estimated normal force differs from the commanded mean by two point three two percent, and the estimated moment differs by four point eight seven percent.

The transient curves, particularly for the moment, do not coincide throughout the complete motion. The percentages therefore refer to the stationary comparison, not the full time history.

This gives supplementary plausibility evidence for the reported wrench levels. It is not an independent validation of sensor accuracy, because the estimate comes from the robot's model-based estimator.

Normal-force plausibility assessment

This assessment compares the implemented translational impedance with its quasi-static spring prediction. The controller held a captured reference, with the compliance centre at the TCP and the null-space torque inactive. A manual displacement was then applied along the surface normal.

In the equation, $K_{p,n}$ is normal translational stiffness. e_p is the position error, and $e_{p,0}$ is its value before the perturbation. Their difference references the calculation to that starting state. Multiplication by n_s^T selects the normal component. The subscript qs means quasi-static prediction.

With a stiffness of one thousand newtons per metre and a measured normal displacement of minus nineteen point six five millimetres, the prediction is minus nineteen point six five newtons.

The commanded mean is minus nineteen point six four newtons, a difference of zero point zero five percent. The model-estimated mean is minus twenty-two point two nine two newtons, thirteen point five percent from the command.

The close spring-to-command agreement supports the plausibility of the implemented translational impedance relation.

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0}).$$

Moment plausibility assessment

The rotational assessment uses the same idea, but the tool is rotated about the first surface tangent while the normal load remains applied.

Here, K_{R,t_1} is rotational stiffness about that tangent. e_R is the orientation-error vector, and $e_{R,0}$ is the error before the rotational perturbation. Multiplication by $t_1^\top$ selects the component about the first tangent.

The starting reference is now the loaded stationary state. Subtracting it removes the moment already present under normal load, so the comparison isolates the response to the additional rotation.

The measured rotation is six point five six five degrees. With rotational stiffness of fifteen newton metres per radian, the spring prediction and the commanded mean are both one point seven one nine newton metres to the reported precision.

The model-estimated mean is one point eight five two newton metres, a difference of seven point seven five percent from the command. This supports the plausibility of the implemented rotational impedance relation.

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0}).$$

End-effector pose representation

This backup gives the mathematical representation of the measured tool pose. The position, p_{EE} , is a three-component vector. The orientation, R_{EE} , is a rotation matrix in $SO(3)$, the set of three-dimensional rotation matrices.

$$p_{EE} \in \mathbb{R}^3, \quad R_{EE} \in SO(3).$$

Position and orientation are expressed relative to the robot base frame. The end-effector frame is attached to the controlled TCP. Together, they define the six-degree-of-freedom pose, collected in the homogeneous transformation.

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}.$$

The controller obtains this measured pose directly from the robot state for feedback.

Null-space kinematics

At a nonsingular configuration, the geometric Jacobian of this seven-joint robot has six independent rows. The rank-nullity relation leaves one independent null-space direction.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1.$$

A joint velocity in that direction gives zero instantaneous tool velocity.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}.$$

The motion generally coordinates several joints. The controller projects the secondary torque contributions into the kinematic null space.

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}.$$

Damping opposes redundant joint velocity. Conditioning actively steers the configuration. The velocity relation is an instantaneous kinematic statement; actual Cartesian position retention was assessed in the pose-hold experiment.

Contact response, normal force and moment

This full comparison shows the three centre positions for the positive-entry-tilt condition. The upper panel gives the angular response, the middle panel the estimated normal force, and the lower panel the estimated interaction moment about the first tangent at the TCP.

The final-second normal-force means are approximately minus eighty-two point nine, minus seventy-nine, and minus seventy-eight newtons at minus forty millimetres, the TCP, and plus forty millimetres. These levels are similar across the three positions, while the interaction moment reverses between the two displaced centres. In these representative traces, the supporting displacement reaches its approximately steady angular response about one point three seconds earlier than the TCP-centred case.

The force and moment are model-based estimates supplied by the robot. This comparison connects the directional angular response to the interaction moment under similar final normal pressing levels.